

Thermodynamic State Vector Stock Conservation Delta Calculator: Sprint 055 Research Report

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Abstract

In complex biogeochemical simulation engines, maintaining strict mass-energy conservation and physical bounds across multi-element cycles (Carbon, Nitrogen, Phosphorus, Water, and Energy) is paramount. Sprint 055 introduces the *Thermodynamic State Vector Stock Conservation Delta Calculator* (`src/thermodynamics/state_validator.ts`). This module isolates the mathematical calculation of expected stock changes (ΔS) derived from boundary flux rates (F) and simulation time steps (Δt). By formally enforcing First Law conservation ($\Delta S = \sum \text{Inflows} - \sum \text{Outflows}$) and Second Law non-negativity bounds ($S_{t+\Delta t} \geq 0$), the validator eliminates silent mass creation/destruction artifacts and guarantees thermodynamic consistency across ecological monad transitions.

Official Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

1 Introduction & Theoretical Foundations

Ecosystem dynamics operate under strict thermodynamic constraints. Within the WEB OF LIFE simulation architecture, matter and energy are modeled as discrete state vectors undergoing continuous transformations governed by boundary fluxes.

1. **First Law of Thermodynamics (Conservation of Mass-Energy):** The net accumulation rate of any stock S_i over a time step Δt must precisely equal incoming fluxes minus outgoing fluxes:

$$\Delta S_i = \left(\sum_j F_{in,j} - \sum_k F_{out,k} \right) \cdot \Delta t \quad (1)$$

2. **Second Law of Thermodynamics (Non-Negativity & Bounds):** Physical stocks cannot hold negative quantities ($S_i(t + \Delta t) \geq 0$). Furthermore, energy transformations must adhere to entropy dissipation gradients, forbidding perpetual creation or unphysical depletion.

2 Architectural Implementation

The `ThermodynamicStateValidator` class provides a pure mathematical utility for computing expected stock deltas without side effects or implicit global state mutations.

$$\begin{aligned}
\text{Net Inflow}(S) &= \sum_{j \in \text{Inputs}} (F_{j,\text{rate}} \cdot \Delta t) \\
\text{Net Outflow}(S) &= \sum_{k \in \text{Outputs}} (F_{k,\text{rate}} \cdot \Delta t) \\
\Delta S &= \text{Net Inflow}(S) - \text{Net Outflow}(S) \\
S_{t+\Delta t} &= S_t + \Delta S \geq 0
\end{aligned} \tag{2}$$

3 Verification and Conclusion

Unit testing suites (`tests/sprint_055.test.ts`) verify that zero-flux equilibria leave state vectors untouched ($\Delta S = 0$), constant positive fluxes scale linearly with Δt , and artificial mass injections trigger immediate thermodynamic violation exceptions.

For ongoing developments, architectural specifications, and full source code, please refer to the official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.